



# Double Shadows at the Lunar Poles

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## Abstract

Topographic depressions within permanently shadowed regions (PSRs) on low-obliquity planetary bodies can be shielded not only from direct illumination, but also from scattered sunlight and thermal emission from nearby sunlit surfaces. We apply an illumination model to high-resolution topography data to produce the first map of double shadows at the lunar poles. The total doubly shadowed area resolved at the 30 m scale is 1.47 km<sup>2</sup> in the northern hemisphere and 5.37 km<sup>2</sup> in the southern hemisphere, predominantly contained within small craters on the floors of large permanently shadowed craters. In total, ~0.04% of PSR area is permanently double shadowed, but the largest double shadows are nearly 600 m across, potentially large enough to be resolved in remote sensing temperature measurements. In the absence of secondary heating sources, double shadows are predicted to be significantly colder than singly shadowed areas and could sequester highly volatile compounds like carbon dioxide. Future in situ investigation of the regions identified in this work can elucidate the origin and distribution of the Moon's most volatile deposits.

*Unified Astronomy Thesaurus concepts:* [The Moon \(1692\)](#); [Planetary polar regions \(1251\)](#); [Lunar surface \(974\)](#); [Surface ices \(2117\)](#)

## 1. Introduction

The polar regions of low-obliquity planetary bodies like the Moon experience a wide range of illumination conditions. In close proximity to areas of near-constant illumination are deep crater interiors where the Sun never shines. Watson et al. (1961) first recognized that temperatures within permanently shadowed regions (PSRs) could be low enough to sequester volatiles like water. In the absence of direct solar heating, PSRs reach temperatures below 100 K, and water transported to polar shadows by volatile-rich impactors (Gibson & Moore 1973; Mandt et al. 2022), volcanic outgassing (Needham & Kring 2017), or produced in situ by solar wind-regolith interactions (Arnold 1979; Hurley et al. 2017) is stable against sublimation for billions of years. These shaded environments hold clues to the origin and distribution of volatiles in the inner solar system and are a prime target for future robotic (Colaprete et al. 2019) and human exploration (Smith et al. 2020).

Previous studies have mapped PSRs through a number of techniques including Earth-based radar (Margot et al. 1999), seasonal imaging (Bussey et al. 1999, 2003), and laser altimetry (Mazarico et al. 2011, 2018). The last of these sources has yielded the most comprehensive and highest-resolution PSR maps to date. Mazarico et al. (2011) developed a method for assessing direct illumination based on the horizon height visible from a point on the lunar surface and identified regions of permanent shadowing poleward of ~80° at a spatial scale of 240 m. PSRs have also been mapped on Mercury (Butler et al. 1993; Chabot et al. 2012) and Ceres (Schorghofer et al. 2016; Platz et al. 2017). Unlike the Moon, PSRs on these bodies have been found to contain extensive areas of radar-bright (Slade et al. 1992; Harmon 2008; Chabot et al. 2013) or reflective (Neumann et al. 2013; Deutsch et al. 2017; Platz et al. 2017) surface

materials indicative of the presence of abundant water ice deposits.

Although shielded from direct solar illumination, PSRs are not completely dark. Shaded crater floors are exposed to scattered sunlight and thermal emission from nearby sunlit surfaces, which differ in intensity from point to point and throughout the lunar year. As a result, illumination (and, consequently, temperature) varies within PSRs. Topographic depressions within PSRs can be doubly shadowed: shielded not only from the Sun but also from secondary illumination sources. The simplest conception of a double shadow is a small crater formed on the floor of a large permanently shadowed crater. The rim of the small crater shields its interior from scattered light and thermal emission from the periodically illuminated rim of the larger crater. These double permanently shadowed regions (DPSRs) are among the coldest places in the inner solar system, with theoretical temperatures approaching 25 K—the minimum possible surface temperature on the Moon (Carruba & Coradini 1999). Such low temperatures would permit cold trapping of not only water but also super-volatiles like CO<sub>2</sub>, CO, N<sub>2</sub>, Ar (Hodges 1980), and organic species found in comets and meteorites (Zhang & Paige 2009; Landis et al. 2022). If delivery rates exceed destruction by solar wind sputtering and micrometeorite gardening (Farrell et al. 2019; Costello et al. 2021) the presence and unique fingerprint of volatiles sequestered in doubly shadowed regions could therefore offer a direct test of the impact hypothesis for the origin of lunar volatiles (Gibson & Moore 1973; Chyba 1987; Greenwood et al. 2011). In contrast to PSRs, a catalog of double shadows does not yet exist in the literature.

The goal of this study is to map single and double PSRs at the lunar poles with high-resolution digital terrain models (DTMs) and characterize how the abundance of double shadows varies with scale. The small size and relative rarity of double shadows has previously made direct study of these regions challenging but unresolved observations indicate that DPSRs do exist although the locations and extent of these areas are unknown. Carbon dioxide cold traps have been inferred in



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spatially contiguous regions over scales of hundreds of meters based on temperature measurements from the Diviner radio-meter instrument (Schorghofer et al. 2021). Additionally, far-infrared emissivity features in Diviner bolometric temperature data have been attributed to anisothermality from subresolution doubly shadowed areas (Sefton-Nash et al. 2019). Using a scattered illumination model, Kloos et al. (2021) estimated that at a scale of 240 m (comparable to the footprint of the Diviner instrument), only  $\sim 0.03\%$  of the PSR area on the Moon is doubly shadowed. These studies support the paradigm that double shadows make up an exceedingly small fraction of the total surface area at the lunar poles. Whether these regions exist more commonly as shallow depressions on rough terrain (Petrov et al. 2003) or as quasicircular regions within simple craters is of particular interest to the study of polar volatiles. If the latter is the case, some double shadows could be large enough to be directly targeted by remote sensing instrumentation or future landed missions.

## 2. Methods

First, we describe the topography data used to assess illumination conditions and discuss the resolution and coverage of these data sets (Section 2.1). Singly shadowed areas are identified based on the elevation of the local topography via the horizon method of Mazarico et al. (2011) (Section 2.2). The resulting PSR map is passed to a raycasting algorithm that determines which permanently shadowed areas are also doubly shadowed based on the elevation and shadowing state of the surrounding topography (Section 2.3).

### 2.1. Data

The Lunar Orbiter Laser Altimeter (LOLA; Smith et al. 2010) has been collecting data for over a decade on board the Lunar Reconnaissance Orbiter (LRO) which operates in a near-polar eccentric orbit with periapsis over the south pole. Publicly available gridded topography maps have a vertical precision of  $\sim 10$  cm. For the primary mapping of single and double shadows in this study, we use polar stereographic DTMs with coverage down to  $75^\circ$  latitude at  $30 \text{ m pxl}^{-1}$  spatial resolution, nearly an order of magnitude higher than that of DTMs used in previous studies of secondary illumination within lunar PSRs (Mazarico et al. 2011; Kloos et al. 2021). Despite restricted areal coverage, higher-resolution  $5 \text{ m pxl}^{-1}$  DTMs ( $>87.5^\circ$  latitude) can also be used to extend our estimates of double shadow abundance to even smaller scales.

### 2.2. Single Shadowing

A point is permanently shadowed if it is never directly illuminated, i.e., no part of the solar disk is ever visible from that location. Mazarico et al. (2011) pioneered the horizon method to map lunar PSRs and polar illumination maps derived from that study are available at a resolution of  $60 \text{ m pxl}^{-1}$  down to  $85^\circ$  latitude. We use a similar approach here to compute direct solar illumination as illustrated in Figure 1.

For each pixel in the region of interest ( $>80^\circ$  latitude, red circle in Figure 1(a)), we start with a gnomonic projection centered on that pixel. Gnomonic map projections preserve line-of-sight angles and centering the projection on each pixel (rather than having a single gnomonic projection centered on the pole) minimizes distortion when reprojecting topography from the original polar stereographic DTM. In the gnomonic

projection, one-dimensional line-of-sight profiles are drawn from the center pixel in 720 azimuthal directions at  $0.5^\circ$  intervals. Profiles extend to a distance of  $d_{\text{max}} \sim 150 \text{ km}$  ( $5^\circ$  along a great circle). The maximum possible distance at which a high-standing topographic feature could be visible above the horizon from a point on the lunar surface is 300 km, but near the poles actual horizon distances tend to be much shorter (Li et al. 2008; Mazarico et al. 2018). Increasing  $d_{\text{max}}$  does not significantly change our results and would further restrict the mapping region. For a  $>75^\circ$  latitude DTM and a maximum horizon distance of 150 km, PSRs can be mapped from the pole to  $80^\circ$  latitude and DPSRs subsequently from the pole to  $85^\circ$  latitude.

Next, we reproject the gnomonic coordinates of each 1D profile to polar stereographic coordinates and bilinearly interpolate the LOLA DTM at those points (Figure 1(b)). The planetary curvature-corrected elevation angle of points along a body,  $\mu_h(d)$ , as viewed from the center pixel is

$$\mu_h(d) = \tan^{-1} \left( \frac{z_0[z(d) - \sqrt{d^2 + z_0^2}]}{d z(d)} \right), \quad (1)$$

where  $z_0$  is the planetary radius at the central pixel,  $d$  is distance along the line-of-sight profile in the gnomonic projection, and  $z(d)$  is the planetary radius of the 1D profile at distance  $d$ . See Appendix for a full derivation of Equation (1). The horizon along each look direction is the point with the greatest elevation angle seen from the center pixel.

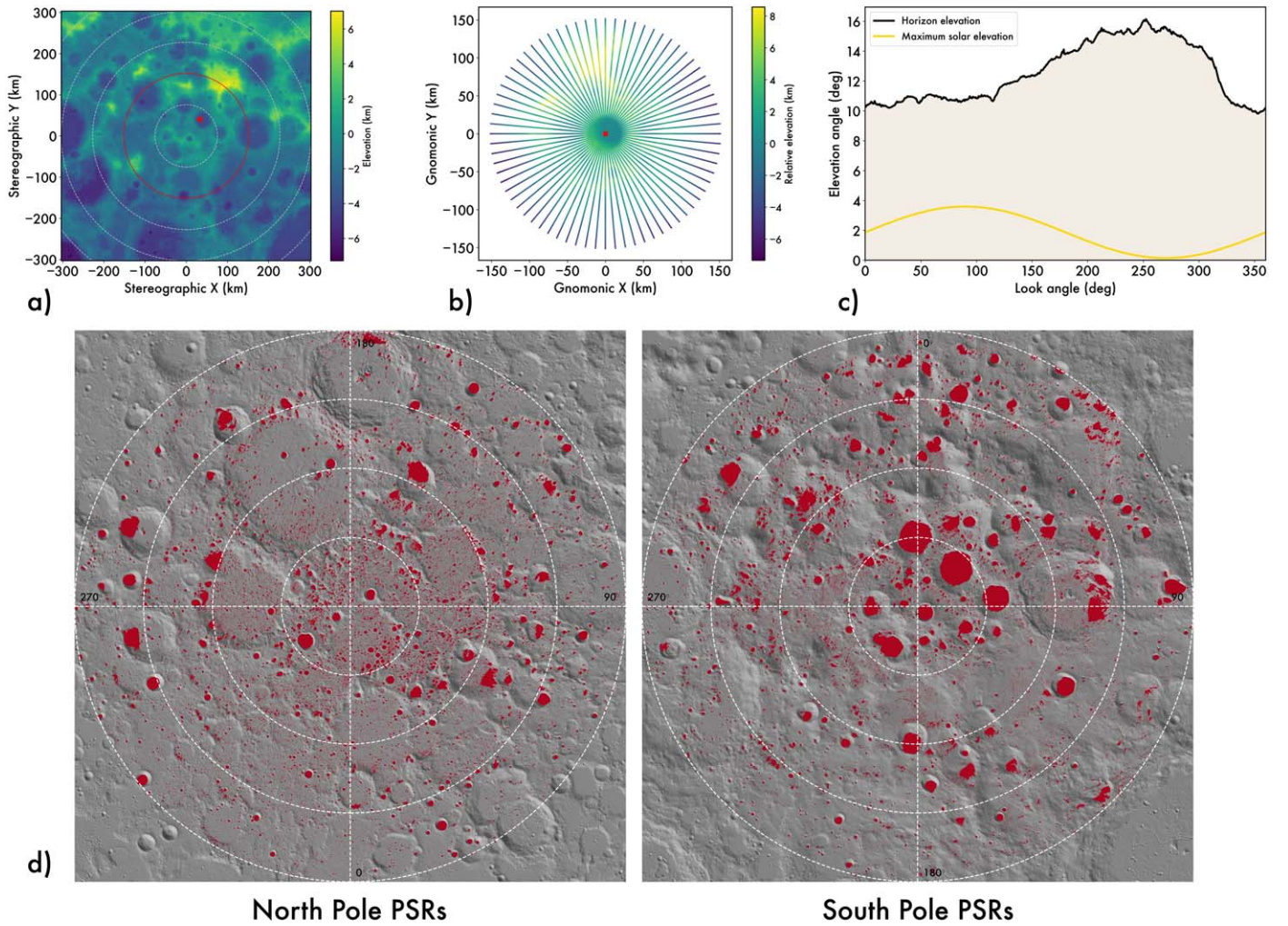
The horizon elevation angle is then compared with the maximum solar elevation angle,  $\mu_\odot$ , in each look direction (Figure 1(c)), which is a function of the latitude of the central pixel,  $\lambda_0$ , the hour angle,  $HA$ , and the lunar obliquity,  $\epsilon$  ( $1^\circ.595$ ; Acton (1996)). The finite size of the solar disk is accounted for by adding the maximum apparent angular radius of the Sun,  $\delta_\odot$  ( $0^\circ.272$ ; Acton (1996)):

$$\mu_\odot = \sin^{-1}(\sin(|\lambda_0|) \sin(\epsilon) + \cos(|\lambda_0|) \cos(\epsilon) \cos(HA)) + \delta_\odot. \quad (2)$$

If the maximum solar elevation angle is less than the horizon elevation angle in every direction, the Sun will never appear above the horizon and therefore the center pixel is permanently shadowed. This process is repeated for every pixel in the region of interest to produce a binary map of permanent shadowing. Pixels that are self-shadowed in a given look direction, i.e., sloping away from that direction, will see a very close and high horizon, eliminating the possibility of direct illumination from that look angle. Figure 1(c) shows a plot of horizon and maximum solar elevation as a function of look direction for a point within Shoemaker crater (red X in Figure 1(a)), a large south polar PSR.

The horizon method offers a simpler alternative to computationally expensive raytracing methods. One advantage of our approach is that no ephemeris is required, as there is no time-dependent element to our model. Pixels that are shadowed will be shadowed for as long as the Moon's obliquity remains fixed at its current value. The Moon's axial tilt has been low enough to sustain polar PSRs for the last  $\sim 2$  Gyr, but earlier in its orbital evolution, may have reached as high as  $77^\circ$  (Sieglar et al. 2011, 2015). Using the Sun's maximum possible elevation rather than its predicted position in the sky over the course of a single lunar orbital cycle yields a slightly more





**Figure 1.** (a)–(c) Horizon method for identifying PSRs in LOLA topography, as described in Section 2.2. (d) Lunar PSRs at 30 m pxl<sup>-1</sup> resolution, shown as red filled regions over hillshaded LOLA topography. Dashed circles indicate latitude intervals of 2°. The reader is referred to the following repository for full resolution shadow maps: O'Brien & Byrne (2022).

conservative estimate of the Moon's PSR area than previous studies, e.g., Mazarico et al. (2011).

### 2.3. Double Shadowing

A point is doubly shadowed if it is permanently shadowed and has no direct line of sight to any nonpermanently shadowed surfaces. For every PSR pixel, we repeat the process to produce curvature-corrected planetary radii profiles centered on the pixel and determine the shadowing state of all points along the profiles using nearest neighbor interpolation of the binary PSR map. Lines are then drawn from the center pixel to every point along each profile. If a line does not intersect the profile between the center pixel and the endpoint, that point is visible from the center pixel. By definition, the horizon is always visible, but intermediate points may be sloping away from the center pixel or obstructed by topographic features. In Figure 2, a representative elevation profile is shown for a point at the bottom of a small crater on the floor of Shoemaker crater (inset in Figure 2). The point is shielded from the partially illuminated wall of the larger crater by the rim of the small crater and every visible point within the horizon is also contained within the large PSR. If every visible surface facet in every direction is permanently shadowed, the center pixel is

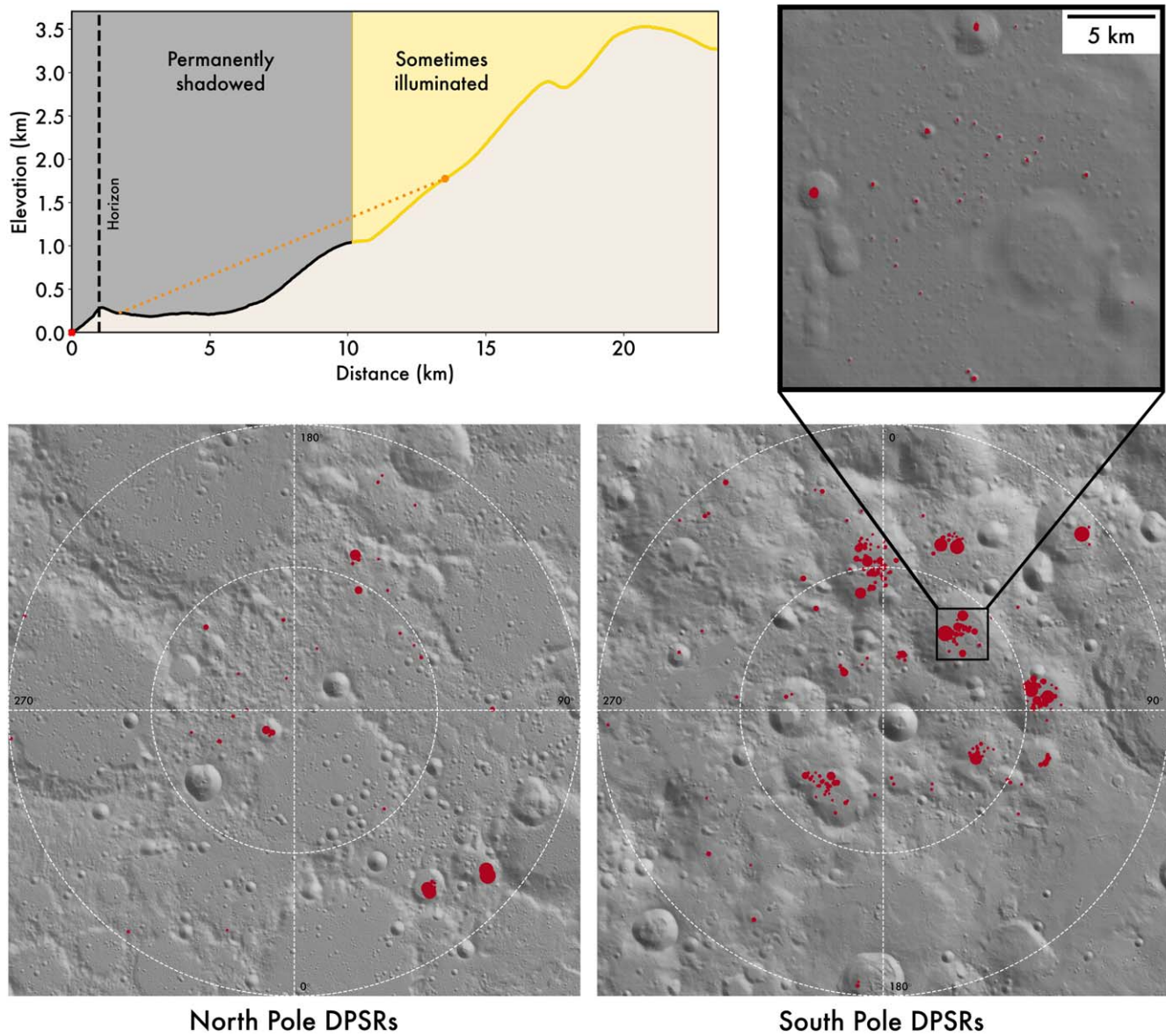
doubly shadowed. This process is repeated for all PSR pixels to produce a second binary map of double shadowing.

A geometric model of double shadowing offers simplicity and speed compared with scattered flux models, at the expense of detailed secondary illumination knowledge. The calculation determines whether a point on the surface is doubly shadowed but provides no information about the energy balance within double shadows. Additional modeling is required to assess whether temperatures within the double shadows identified in this study approach the predicted 25 K minimum (Carruba & Coradini 1999).

## 3. Results

### 3.1. Single Shadowing

We first validated our direct illumination model by running it on the 60 m pxl<sup>-1</sup> LOLA DTMs and comparing the results with the >85° PSR map from Mazarico et al. (2011). Our method produces a nearly identical map of PSR locations and a total PSR area that is 0.40% lower in the north and 0.38% lower in the south, demonstrating good agreement with previous studies at that scale despite our more conservative definition of permanent shadowing. Having validated our method, we then used the higher-resolution LOLA DTM (30 m pxl<sup>-1</sup>) and



**Figure 2.** Upper left: a location is doubly shadowed if every visible surface facet along every line of sight is permanently shadowed. Bottom: map with dots showing locations of double shadows identified at  $30 \text{ m pxl}^{-1}$  (larger than 5 contiguous pixels). Dashed circles indicate latitude intervals of  $2.5^\circ$ . Dot size is exaggerated but depicts relative area of each DPSR. Inset (upper right) shows resolved double shadows within Shoemaker crater (DPSR area not exaggerated here).

expanded the mapping area to identify all PSRs poleward of  $80^\circ$  latitude. The resulting PSR map (Figure 1(d)) is qualitatively similar to previous studies with a slight refinement at the edges of large PSRs. The primary difference is the identification of numerous small PSRs that could not be identified in lower-resolution topography data, consistent with past studies that demonstrated the total resolved shadow area increases as spatial scale decreases (Smith et al. 2017; Hayne et al. 2021). To avoid potentially spurious features from artifacts in the DTM, shadows smaller than 5 contiguous pixels, i.e., connected by sides or corners, have been excluded from these totals. As a result of the increased resolution, we obtain a substantially higher value for the total PSR area poleward of  $80^\circ$  latitude compared with previously published metrics:  $1.80 \times 10^4 \text{ km}^2$  in the north and  $2.09 \times 10^4 \text{ km}^2$  in the south (6.24% and 7.26% of the total surface area, respectively). For comparison, Mazarico et al. (2011) reported values of  $1.29 \times 10^4 \text{ km}^2$  and  $1.61 \times 10^4 \text{ km}^2$  poleward of  $\sim 80^\circ$  latitude

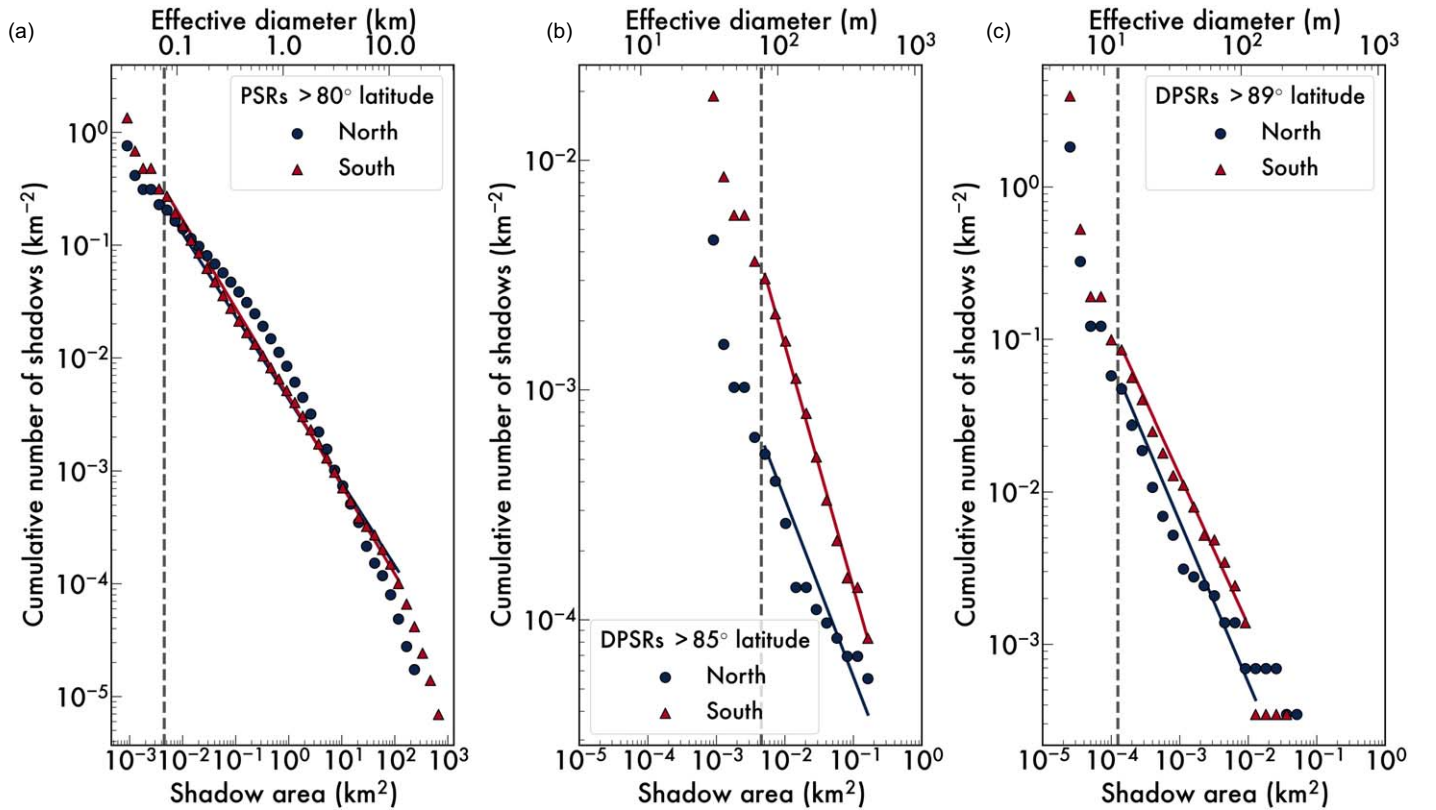
(4.7% and 5.8% of the total area in their similar region of interest).

Figure 3(a) shows the cumulative size-frequency distribution (SFD) of lunar PSRs. Small shadows are more numerous than large ones and this trend continues down to the  $30 \text{ m pxl}^{-1}$  resolution limit at shadow areas of  $4500 \text{ m}^2$  (five contiguous pixels) or, equivalently, an effective shadow diameter of 76 m, where effective diameter is defined as the diameter of a circle with the same area as the shadow. Over multiple orders of magnitude, the PSR size-frequency distribution can be approximated by a power law,

$$N(>A) = C A^{-b}, \quad (3)$$

where  $A$  is shadow area. The coefficient,  $C$ , is constrained so that the SFD matches the observed cumulative number density resolved at the minimum shadow area of 5 pixels. A least-squares fit to the power law slope yields  $b_n = 0.74$  for the north and  $b_s = 0.78$  for the south. As can be seen in Figure 3(a), a





**Figure 3.** Size-frequency distribution of 30 m pxl<sup>-1</sup> PSRs (a) and DPSRs (b) and 5 m pxl<sup>-1</sup> DPSRs (c). Dashed lines show the resolution limit at 5 contiguous pixels.

single power law does not precisely capture the structure inherent in the PSR size-frequency distributions and there is a clear hemispherical dichotomy in shadow abundance that has been noted previously (Hayne et al. 2021). The northern hemisphere SFD appears to change slope at  $\sim 0.1 \text{ km}^2$  and for both hemispheres there is a rolloff at shadows larger than  $\sim 100 \text{ km}^2$ . However, the power law approximation offers a useful heuristic to estimate how shadow abundance varies with scale.

### 3.2. Double Shadowing

We next mapped all double shadows poleward of  $85^\circ$  latitude using the 30 m pxl<sup>-1</sup> LOLA DTM and PSR map (Figure 2). The total area of DPSRs resolved at 30 m pxl<sup>-1</sup> is  $1.47 \text{ km}^2$  in the north and  $5.37 \text{ km}^2$  in the south. Double shadows smaller than five contiguous pixels have been excluded from these totals. These areas represent, respectively, 0.018% and 0.055% of the permanently shadowed area in the same latitude region. Using a lower-resolution 240 m pxl<sup>-1</sup> DTM, Kloos et al. (2021) computed the incident flux of secondary illumination within PSRs and obtained a slightly lower value for the DPSR area fraction, with 0.03% of PSR elements receiving no scattered sunlight over one lunar year. Overall,  $\sim 0.005\%$  of the total surface area in the Moon's polar regions lies in double shadow. As shown in Figure 2, double shadows tend to be concentrated on the flat floors of large craters. The three south polar PSRs in Haworth, Shoemaker, and Faustini host numerous small craters with doubly shadowed interiors that are fully resolved in the 30 m pxl<sup>-1</sup> LOLA DTM. The largest of these shadows (leftmost shadow in inset of Figure 2) is nearly 600 m across. It should be noted that

not all small craters embedded in PSRs are doubly shadowed. Those not identified as double shadows by our model are either too shallow or too wide to be fully shielded from the partially illuminated wall of the larger crater or other distant high-standing topographic features. The incident flux within nondoubly shadowed craters may be lower than on surrounding flat PSR terrain, but the very coldest temperatures will still be found within DPSRs. Table 1 contains a catalog of the location and size of the largest doubly shadowed regions.

Using Diviner temperature measurements gridded to a resolution of 500 m pxl<sup>-1</sup>, Schorghofer et al. (2021) computed sublimation rates at the lunar south pole and identified small spatially contiguous regions of CO<sub>2</sub> ice stability. Some of the PSRs found to contain CO<sub>2</sub> cold traps, e.g., Haworth, Shoemaker, and Faustini, contain numerous double shadows. However, there are also PSRs containing double shadows that were not found to contain CO<sub>2</sub> ice stability zones at the 500 meter scale and the large cold trap in de Gerlache Crater does not correspond to high double shadow abundance (Figure 2). At present, the spatial resolution of the Diviner data set is too low to facilitate direct comparison of CO<sub>2</sub> cold trap locations with the map of DPSRs. A determination of temperature and volatile stability within double shadows is also beyond the scope of the present study.

As with single shadows, small double shadows are more common than large ones and the DPSR size-frequency distributions (Figure 3(b)) have slopes  $b_n = 0.77$  for the north and  $b_s = 1.06$  for the south. LOLA gridded data resolution is better than 30 m pxl<sup>-1</sup> near the poles, allowing us to assess whether the double shadow abundance trend shown in Figure 3(b) continues to scales relevant for landed missions. The 5 m pxl<sup>-1</sup> LOLA DTMs covering  $>87.5^\circ$  latitude can

**Table 1**

Size and Location of the Largest Doubly Shadowed Regions on the Moon

| Area<br>(km <sup>2</sup> ) | $D_{\text{eff}}$ (m) | Latitude (°) | Longitude (°) | Lunar Features                      |
|----------------------------|----------------------|--------------|---------------|-------------------------------------|
| 0.274                      | 590.219              | 85.552       | 49.426        | Fibiger, Byrd D,<br>Nansen F Crater |
| 0.262                      | 577.461              | -88.275      | 38.992        | Shoemaker Crater                    |
| 0.242                      | 555.204              | -85.362      | 48.430        | Nobile Crater                       |
| 0.220                      | 529.858              | 85.636       | 50.220        | Fibiger, Byrd D,<br>Nansen F Crater |
| 0.213                      | 521.135              | -86.870      | 24.335        | Haworth, Shoemaker<br>Crater        |
| 0.201                      | 505.509              | -87.390      | 82.312        | Faustini Crater                     |
| 0.187                      | 488.211              | 86.035       | 36.598        | Fibiger Crater                      |
| 0.182                      | 481.118              | -86.940      | 19.223        | Haworth, Shoemaker<br>Crater        |
| 0.176                      | 473.919              | -87.114      | 85.381        | Faustini Crater                     |
| 0.167                      | 461.671              | 86.113       | 36.769        | Fibiger Crater                      |
| 0.152                      | 440.068              | -88.171      | 117.626       | Slater Crater                       |
| 0.126                      | 400.535              | -87.915      | -10.942       | Haworth Crater                      |
| 0.121                      | 391.858              | -87.381      | -6.105        | Haworth Crater                      |
| 0.116                      | 384.478              | 87.086       | 158.568       | Bosch Crater                        |
| 0.114                      | 381.486              | -87.841      | 40.078        | Shoemaker Crater                    |
| 0.111                      | 375.430              | -87.436      | 80.655        | Faustini Crater                     |
| 0.076                      | 312.094              | -87.310      | 86.333        | Faustini Crater                     |
| 0.071                      | 300.878              | -88.531      | -141.567      | Sverdrup Crater                     |
| 0.070                      | 298.967              | -88.049      | 41.710        | Shoemaker Crater                    |
| 0.068                      | 295.109              | 89.398       | -54.699       | Hinshelwood Crater                  |
| 0.063                      | 283.221              | -89.053      | -45.939       | de Gerlache, Haworth<br>Crater      |
| 0.061                      | 279.146              | -87.138      | -8.665        | Haworth Crater                      |
| 0.058                      | 270.811              | 87.621       | 151.865       | Bosch Crater                        |

extend the DPSR resolution limit down to shadow sizes of 125 m<sup>2</sup> (5 pixels), with some caveats. It is computationally prohibitive, though not strictly necessary, to recompute single shadowing at this scale. By upsampling the 30 m pxl<sup>-1</sup> PSR map to the same resolution as the 5 m pxl<sup>-1</sup> DTM, we may exclude some small undetected PSRs that could host double shadows. However, double shadows tend to be concentrated on the floors of large craters, i.e., deep within large PSRs. Incompletely resolving the edges of those PSRs or the shadowing state of distant topography does not affect whether a small embedded crater is doubly shadowed, as the horizon typically occurs at the well-resolved rim of the small crater (Figure 2).

The restricted areal coverage of the 5 m pxl<sup>-1</sup> DTMs prohibits checking a maximum horizon distance of  $d_{\text{max}} = 150$  km (5°), the value used in the 30 m pxl<sup>-1</sup> mapping to capture distant high-standing topographic features. We tested the effects of degrading  $d_{\text{max}}$  on southern double shadows and found that at 30 m pxl<sup>-1</sup>, reducing the maximum horizon distance from 5° to 1° increased the total DPSR area in the region by ~5%. A small number of pixels that were not doubly shadowed under on the full 5° horizon calculation, due to some distant topographic feature being both visible and not permanently shadowed, flipped to be doubly shadowed when a shorter horizon distance was checked. Therefore, using  $d_{\text{max}} = 1°$  introduces an error of at most a few percent to the total PSR area identified as being doubly shadowed.

Figure 3(c) shows the size-frequency distribution of double shadows mapped poleward of 89° latitude at 5 m pxl<sup>-1</sup> resolution. The best fit power laws have slopes  $b_n = 1.06$  for the north and  $b_s = 0.98$  for the south, comparable to the slope

of the combined north and south DPSR size-frequency distribution at 30 m pxl<sup>-1</sup>. The abundance of double shadows appears to follow a consistent trend from the largest DPSRs, almost 600 m in effective diameter, down to the 5 m pxl<sup>-1</sup> resolution limit at effective shadow diameters of ~13 m. Application of this trend to all sizes predicts that poleward of 85° latitude, double shadows larger than 1 cm<sup>2</sup> (the scale at which lateral heat conduction eliminates cold traps; Hayne et al. 2021) comprise a total surface area of 1.74 km<sup>2</sup> in the north and 44.65 km<sup>2</sup> in the south. It should be noted that this minimum cold trap scale was determined for PSRs and the temperature gradient between sunlit and PSR terrain is much greater than between PSR and DPSR terrain. Small shadows, though more abundant, do not dominate the total DPSR area, e.g., ~15% of the Moon's doubly shadowed area is resolved at 30 m pxl<sup>-1</sup>. The significant hemispherical discrepancy in DPSR area is due to the difference in SFD slope between the two populations which in turn is likely a result of the discrepancy in PSR abundance. The southern hemisphere, with its excess of large PSRs, has more permanently shadowed area where small depressions can be doubly shadowed.

#### 4. Conclusions

In this work, we have presented the first map of double permanently shadowed regions at the lunar poles, as well as an improved resolution PSR map. We find that some double shadows are large enough to appear in orbital thermal and reflectance measurements, potentially facilitating analysis of the temperature and volatile content of these ultra-cold features. However, the resolution of currently available remote sensing data sets presents a challenge to determining DPSR temperatures. For example, the surface footprint of the Diviner radiometer instrument is 140 × 400 m (Williams et al. 2019), or 0.056 km<sup>2</sup>. We find 23 DPSRs (seven in the north, 16 in the south) that have areas larger than a single Diviner pixel, but none are larger than five Diviner pixels in size. The size and location of these double shadows are given in Table 1. Though it is presently difficult to resolve such small features in the Diviner data set, the largest double shadows constitute compelling targets for future orbital and in situ measurements.

As with PSRs, small double shadows are more abundant than large ones and, even at 5 m pxl<sup>-1</sup>, many DPSRs remain unresolved. However, the consistent power law trend in double shadow abundance over multiple orders of magnitude in shadow area (Figure 3) allows us to make predictions about the abundance DPSRs at subresolution scales. Hayne et al. (2021) found that lateral heat conduction becomes an important factor in warming cold traps at scales of less than ~1 cm. Extrapolation of our results to this minimum scale suggests that the total surface area in double shadows large enough to host ultra-cold traps is ~46 km<sup>2</sup>.

Despite their scarcity, DPSRs have now been conclusively identified with advancements in spatial resolution from laser altimetry-derived topographic data sets. If double shadows are as cold as predicted from thermal modeling (Carruba & Coradini 1999), they could hold clues to the distribution and abundance of the Moon's most thermally unstable volatiles. Having determined their location and areal extent, these extreme illumination environments warrant further study from existing orbital data sets and as targets for future in situ missions.

The LOLA digital terrain models used in this study are publicly available from the Planetary Data System at <http://imbrium.mit.edu/>. The Mazarico et al. (2011) reference PSR maps were also obtained from the LOLA PDS node (<http://imbrium.mit.edu/EXTRAS/ILLUMINATION/>). Software for illumination modeling and analysis can be accessed at <http://github.com/pcobrien/DoubleShadows/>. All shadow maps produced in this work are stored at the following repository: <https://doi.org/10.5281/zenodo.7250966>.

### Appendix Derivation of Curvature-corrected Horizon Elevation Angle

A point,  $P_1$ , on the surface of a body, has radius,  $R_1$ . A distant point,  $P_2$ , with radius,  $R_2$ , is separated from  $P_1$  by an angle,  $\theta$ .  $P_1$ ,  $P_2$  and the center of the body define a plane (Figure 4). In this plane, the body center has position (0, 0),  $P_1$  has position (0,  $R_1$ ), and  $P_2$  has position ( $R_2 \sin \theta$ ,  $R_2 \cos \theta$ ). A line connecting  $P_1$  and  $P_2$  (dotted line in Figure 4) has slope

$$\frac{\Delta y}{\Delta x} = \frac{R_2 \cos \theta - R_1}{R_2 \sin \theta}. \quad (\text{A1})$$

The slope of this line represents the elevation angle,  $\mu$ , of  $P_2$  relative to  $P_1$ . If the elevations of the points are known, all that remains is to determine the value of  $\theta$ , which is a function of the distance between the two points. A gnomonic map projection preserves viewing angles and is therefore well-suited to computing the horizon elevation of line-of-sight elevation profiles. The plane of the gnomonic projection is

shown as a horizontal dashed line in Figure 4. The angle,  $\theta$ , can be expressed as a function of the distance,  $d$ , between  $P_1$  and  $P_2$  in the gnomonic projection:

$$\theta = \tan^{-1} \left( \frac{d}{R_1} \right). \quad (\text{A2})$$

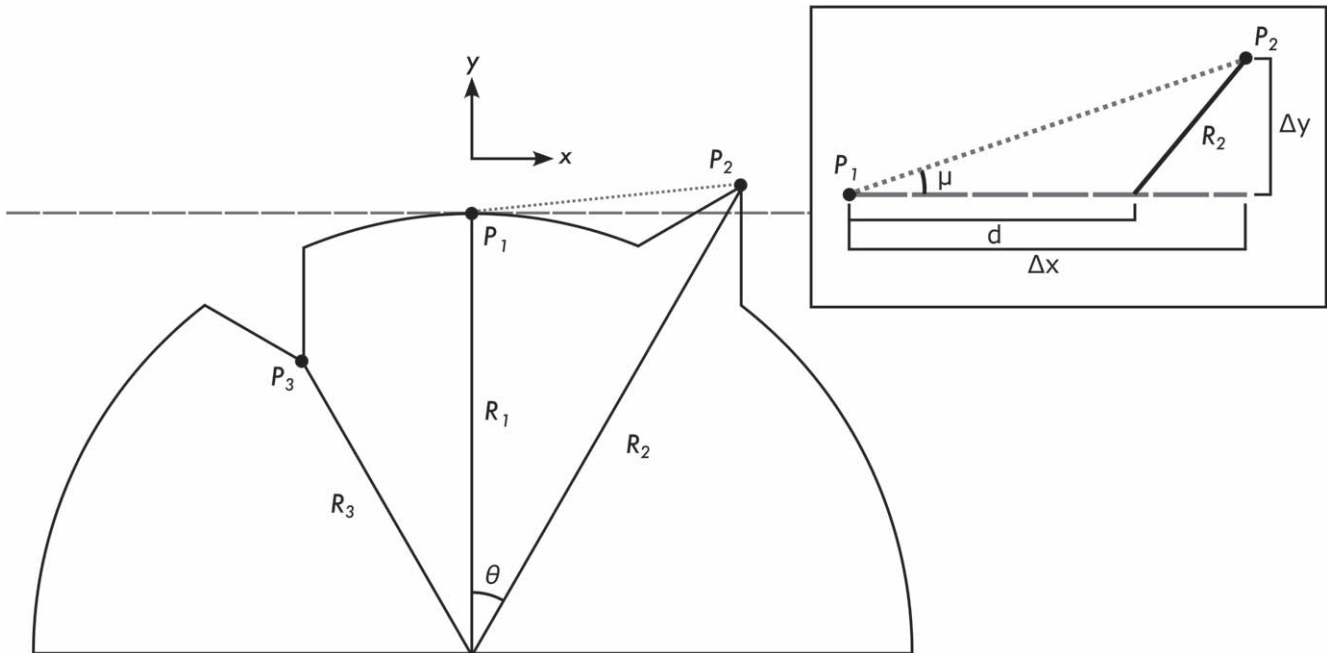
Substituting this expression into Equation (A1), we obtain the elevation angle of  $P_2$  as seen from  $P_1$ :

$$\mu = \tan^{-1} \left( \frac{\Delta y}{\Delta x} \right) = \tan^{-1} \left( \frac{R_1}{d} - \frac{R_1}{R_2 \sin \left( \tan^{-1} \left( \frac{d}{R_1} \right) \right)} \right). \quad (\text{A3})$$

Because  $d$ ,  $R_1$ , and  $R_2$  are strictly positive, this can be further simplified to

$$\mu = \tan^{-1} \left( \frac{R_1(R_2 - \sqrt{d^2 + R_1^2})}{dR_2} \right). \quad (\text{A4})$$

As a sanity check, consider the case where  $P_1$  has a radius equal to the lunar reference radius,  $R_1 = 1737.4$  km. Then, at a distance of 150 km in the gnomonic projection (equivalent to 149.6 km measured along the surface of a sphere of radius  $R_1$ ),  $P_2$  will have a positive elevation angle only if it has a radius 6 km greater than that at  $P_1$ . Conversely, if a point has a smaller radius than  $P_1$ , the elevation angle,  $\mu$ , will be negative regardless of the distance between the points (e.g., point  $P_3$  in Figure 4).



**Figure 4.** Schematic of method to compute the elevation angle of distant topography ( $P_2$ ) on a planetary body as seen from point  $P_1$ .

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